Hebbian Learning

and long-term synaptic plasticity

Computational Neuroscience

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mrule

Learning outcomes:

- Long-term synaptic plasticity
  - Hebbian leaning
  - Supervised vs unsupervised
  - NMDA receptors, STDP
  - Stability: BCM, Oja's Rules

Long-term synaptic plasticity

Long-term synaptic plasticity is a (activity-dependent) semi-permanent change in the strength of the connection from one neuron to another.

- Thought to mediate long-term memory (in contrast to STP)
- Much slower timescale than electrical dynamics.
- Long-term synaptic
  - increases → potentiation (LTP)
  - decreases → depression (LTD)

Hebbian plasticity

"Let us assume that the persistence or repetition of a reverberatory activity (or "trace") tends to induce lasting cellular changes that add to its stability. ... When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased."

— *Donald Olding Hebb (1949)*

"Neurons that fire together wire together."

— *Carla Shatz (1992)*

Supervised, Unsupervised, Something else?

Supervised learning

- Target output value y^* is known for each input $\boldsymbol{x}$
- Tune parameters to make system output y closer to y^*
- In this course: Cerebellum, Perceptron

Unsupervised learning

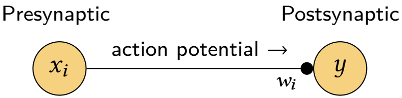
- Only inputs $\boldsymbol{x}$ are provided (no target)
- Learning rules typical learn a model of the structure of $\boldsymbol{x}$
- In this course: Hippocampus area CA3, Hopfield

Is Hebbian learning $\Delta w_i \propto y x_i$ supervised or unsupervised?

- Sometimes conceptually blurry (c.f. Rule, O'Leary 2022)

(w_i is the synaptic weight for presynaptic input x_i onto postsynaptic cell with output y)

Hebbian plasticity



Hebbian rule

$$\frac{1}{\eta}\Delta w_i = y x_i$$

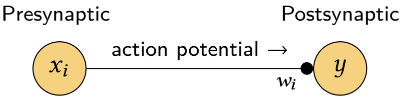
w_i: Synapse from presynaptic input x_i

y: Postsynaptic neuronal output, usually of form $y = \phi(\mathbf{w}^\top \boldsymbol{x})$

η: Learning rate

Unstable: w and hence y grow without bound.

Hebbian plasticity



Continuous time, averaged rule ("gradient flow") over a batch of training pairs $\{\boldsymbol{x}, y\}$, (using $\eta = 1$ for simplicity):

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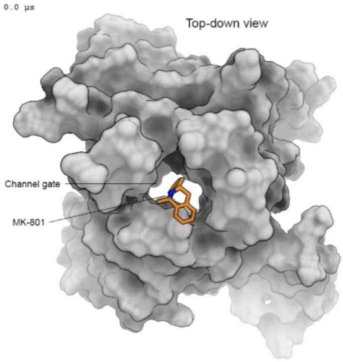
NMDA receptors (NMDARs) can detect correlations between presynaptic and postsynaptic activity, and trigger signals that allow for Hebbian plasticity.

NMDA receptors: coincidence detector



[*Song et al, Nature (2018)*]

NMDA receptor (NMDAR)

- Ionotropic glutamate receptor
- Mainly Ca^{2+} (some Na^{+}, K^{+})

Long-term synaptic plasticity, learning, memory

Affected by common drugs (PCP, alcohol, ketamine, nitrous oxide)

Magnesium (Mg^{2+}) block

Rest ~ -70 mV:

- Extracellular Mg^{2+} binds $\rightarrow$ blocks pore

Depolarisation:

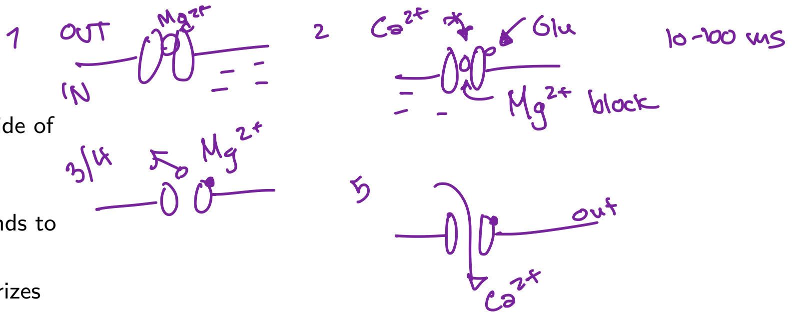
- $\sim -50 - -30$ mV, Mg^{2+} leaves
- $\rightarrow$ NMDAR can open if glutamate binds

Depolarisation + Glutamate $\rightarrow$ open

NMDA is a human-made chemical that selectively activates the receptor; Here, MK-801 = Dizocilpine, an NMDA pore blocker

Draw:

1. Mg^{2+} stuck on the exterior side of the NMDAR pore at rest
2. A barrage of excitatory input releases glutamate, which binds to postsynaptic NMDARs
3. The postsynaptic cell depolarizes (either backpropagating AP or dendritic potential)
4. Depolarisation removes Mg^{2+} block
5. NMDARs with previously bound glutamate allow Ca^{2+} into the cell
6. Ca^{2+} ions trigger lasting chemical changes (plasticity)



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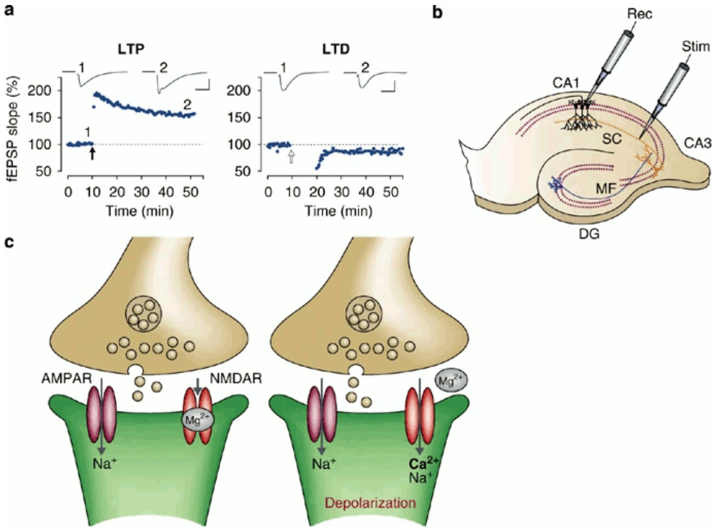
Ca^{2+} binds to stuff



NMDAR-dependent LTP and LTD at hippocampal CA1 synapses, Citri and Malenka (2007)

Spike-Timing-Dependent Plasticity (STDP) is a form of Hebbian plasticity that is sensitive to *temporal* correlations.

Spike-Timing-Dependent Plasticity (STDP)

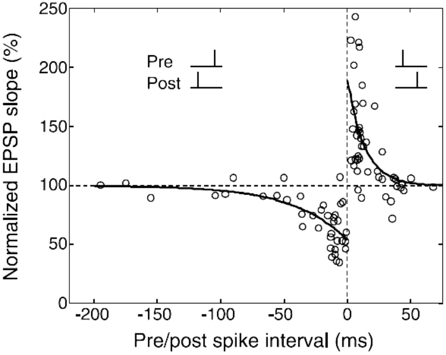
Causal flavour of Hebbian learning
If presynaptic spike A happened just...

- **before** postsynaptic spike B,
  ...A could have caused B
- **after** postsynaptic spike B,
  ...A could not have caused B

Classic STDP:

- Pre-before-post → LTP
- Post-before-pre → LTD

STDP's existence implies that synapses can detect millisecond-level differences in spike timing when deciding whether to strengthen or weaken.



[Dan & Poo (2004)]

Competitive Hebbian learning via STDP

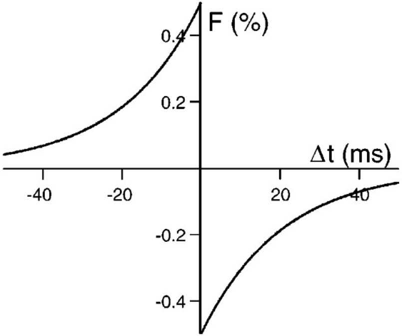
A simple computational model of STDP demonstrated that it can induce competition between the inputs; The group of synaptic inputs with the strongest correlations 'wins'.

STDP = Hebbian + temporal filter $F(t_{\text{pre}} - t_{\text{post}})$:

$$\dot{w}_i \propto \langle x_i(t) \cdot [F * y](t) \rangle_t$$

where $y(t)$ is the output spike train and $x_i(t)$ is the input spike train for synapse with weight w_i.

Simple picture *slightly* controversial: Negative lobe may be from homeostatic changes?



Song, Miller & Abbott, Nat Neurosci (2001)

$$F(\Delta t) = \begin{cases} A_+ \exp(\Delta t/\tau_+) & \text{if } \Delta t < 0 \\ -A_- \exp(-\Delta t/\tau_-) & \text{if } \Delta t \geq 0 \end{cases}$$

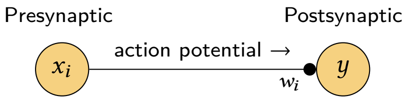
Pause.

Switch slide decks to briefly review gradient descent.

Did you go through the slides?

Resume.

Hebbian plasticity



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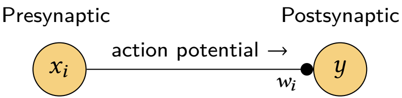
Hebbian plasticity



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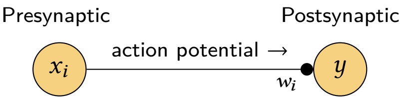

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Is this stable?

Unstable: w and hence y grow without bound.

Is Hebbian learning stable?

To check stability

- When is $\Delta \mathbf{w}$ or $\dot{\mathbf{w}}$ zero? (fixed points)
- Are these fixed points stable (do small perturbations return to them?)
- Does the system globally contract (i.e. pull all states toward) these fixed points?

We'll focus on averaged gradient flow learning rules

- Average possible weight updates over the training data before applying
- Take the continuous-time limit to obtain an ODE for weight evolution
- This reduces stability analysis to checking for stable fixed points in an ODE

Is Hebbian learning stable?

Is this stable?

$$\dot{\mathbf{w}} = \langle y \boldsymbol{x} \rangle$$

Check $\dot{w}_i = 0$:

$$\langle y \boldsymbol{x} \rangle = 0$$

The above is only true when the neuron's output y is identically zero or uncorrelated with inputs $\boldsymbol{x}$? That doesn't seem very useful?

Indeed, Hebbian learning is unstable on its own (synaptic weights keep increasing)

Is Hebbian learning stable?

Is Hebbian learning unstable?

- Technically yes, but it was always understood that “fire together, wire together” would be accompanied by some sort of stabilising biological process. The naïve “unstable” Hebbian rule is just a rhetorical device to introduce the question:

What processes, specifically, ensure stability, and how can we model them?

- Tweaks to make the learning rule inherently stable?
- Explicit **homeostatic processes**?
  - E.g. capping the total number of AMAPRs available to the cell

We will now consider two example stablized Hebbian learning rules: BCM and Oja's rule

Adapted from B Mizusaki

Bienenstock–Cooper–Munro (BCM) rule

Add a multiplicative term to decrease synaptic weight for low levels of postsynaptic activity

$$\begin{aligned} \dot{w} &= \langle y\boldsymbol{x} \cdot (\underline{y - \theta}) \rangle \\ \tau_\theta \dot{\theta} &= \langle y^2 \rangle - \theta \end{aligned} \tag{1}$$

$y - \langle y^2 \rangle$

$\theta \sim \langle y^2 \rangle$

Typically $\tau_\theta \ll 1$ so that $\theta \sim \langle y^2 \rangle$ and we can view (1) as

$$\dot{w} = \langle y\boldsymbol{x} \cdot (y - \langle y^2 \rangle) \rangle = \langle \boldsymbol{x} y^2 \rangle - \langle \boldsymbol{x} y \rangle \cdot \langle y^2 \rangle$$

Adaptive activity-dependent threshold θ stabilises plasticity and hence activity. This makes synapses compete: Inputs that drive larger y are potentiated at the expense of inputs that recruit a weak response.

The full BCM rule is more abstract (general) than the example on the previous slide, but we don't need the details in this course.

(pause for notation aside)

Linear neurons

- Analysing $\dot{\mathbf{w}} = \langle y\boldsymbol{x}\rangle$ further requires defining what y actually is. Today we will look at unsupervised learning in a *linear* neuron, $y = \mathbf{w}^\top \boldsymbol{x}$.

Averaged continuous-time learning rules

- Consider the gradient flow averaged Hebbian rule for a linear neuron $y = \mathbf{w}^\top \boldsymbol{x}$

$$\dot{\mathbf{w}} = \langle y\boldsymbol{x}\rangle = \langle(\mathbf{w}^\top \boldsymbol{x})\boldsymbol{x}\rangle = \langle \boldsymbol{x}\boldsymbol{x}^\top\rangle \mathbf{w} = \boldsymbol{Q}\mathbf{w}$$

- $\boldsymbol{Q} = \Sigma_x + \boldsymbol{\mu}_x \boldsymbol{\mu}_x^\top$ is the second moment of the input data distribution, where
  - $\boldsymbol{\mu}_x = \langle \boldsymbol{x}\rangle$ is the mean (average) input value
  - $\Sigma_x = \mathrm{Cov}(\boldsymbol{x}) = \langle(\boldsymbol{x} - \boldsymbol{\mu}_x)(\boldsymbol{x} - \boldsymbol{\mu}_x)^\top\rangle$ is the covariance matrix for $\boldsymbol{x}$

Weight decay ~ multiplicative scaling

- If $\mathbf{w} = \{w_1, w_2, \ldots, w_n\}^\top$ is a weight vector, the ODE $\dot{\mathbf{w}} = -\lambda \mathbf{w}$ has an exponential solution for the initial value problem $\mathbf{w}(t) = w_0$: $\mathbf{w}(t) = \exp(-\lambda t)\mathbf{w}_0$. For a step Δ_t, this scales all the weights down equally by factor $\alpha = \exp(-\lambda \Delta_t) < 1$.

The following slides assume an averaged rule and a linear neuron, and use notation like $\boldsymbol{Q}, \Sigma_x, \boldsymbol{\mu}_x$.

(end notation aside)

Add homeostasis?

The amount of “stuff” to make synapses is $\approx$ constant

Simple model? Fixed weight-vector length

$$\|\mathbf{w}\|_2^2 = w_1^2 + w_2^2 + \cdots + w_n^2 = 1$$

Monitor the homeostatic error $1 - \|\mathbf{w}\|_2^2$ and add a weight decay/growth term to regulate it?

$$\begin{aligned} \dot{\mathbf{w}} &= Q\mathbf{w} - \lambda \mathbf{w} \\ \dot{\lambda} &= \|\mathbf{w}\|_2^2 - 1 \end{aligned} \tag{2}$$

Oja’s rule achieves the same result, implicitly.

Oja's Rule (1982)

Oja's Rule

$$\dot{\mathbf{w}} = \langle y\boldsymbol{x} \rangle - \langle y^2 \rangle \mathbf{w}$$

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$$\dot{\mathbf{w}} = \langle y\boldsymbol{x}\rangle - \langle y^2\rangle\mathbf{w}$$

With a linear neuron $y = \mathbf{w}^\top \boldsymbol{x}$

$$\begin{aligned}\dot{\mathbf{w}} &= \langle(\mathbf{w}^\top \boldsymbol{x})\boldsymbol{x}\rangle - \langle(\mathbf{w}^\top \boldsymbol{x})^2\rangle \cdot \mathbf{w} \\ &= \langle \boldsymbol{x}\boldsymbol{x}^\top\rangle\mathbf{w} - (\mathbf{w}^\top\langle \boldsymbol{x}\boldsymbol{x}^\top\rangle\mathbf{w}) \cdot \mathbf{w} \\ &= \boldsymbol{Q}\mathbf{w} - \underbrace{(\mathbf{w}^\top \boldsymbol{Q}\mathbf{w})}_{\lambda} \cdot \mathbf{w}\end{aligned}$$

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Equilibrium at $\dot{\mathbf{w}} = 0$ implies $\boxed{\boldsymbol{Q}\mathbf{w} = \lambda\mathbf{w}}$

- This equation is the definition of an eigenvector for matrix $\boldsymbol{Q}$: there's a fixed point whenever $\mathbf{w}$ aligns with an eigenvector of $\boldsymbol{Q} = \langle\boldsymbol{x}\boldsymbol{x}^\top\rangle$.

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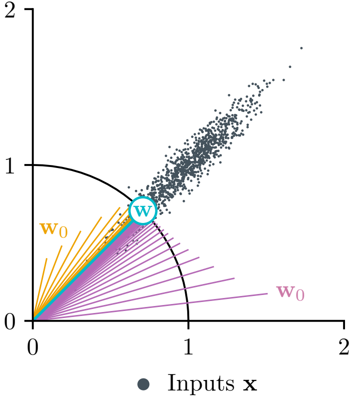
So, we know the direction of $\mathbf{w}$ — what about its length?

- $\mathbf{w}$ satisfies $\mathbf{w}^\top\boldsymbol{Q}\mathbf{w} = \lambda$.
- Since $\boldsymbol{Q}\mathbf{w} = \lambda\mathbf{w}$, we have $\mathbf{w}^\top\lambda\mathbf{w} = \lambda \Rightarrow \mathbf{w}$ is unit length: $\mathbf{w}^\top\mathbf{w} = \|\mathbf{w}\|_2^2 = 1.$

Weight $\mathbf{w}$ evolution of Oja's rule for two initial conditions



Oja's Rule

Oja's Rule with a linear neuron $y = \mathbf{w}^\top \boldsymbol{x}$

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For a linear neuron, Oja's rule learns the same weights as Hebbian + homeostasis in Eqn. (2). Fixed points for $\mathbf{w}$ are unit vectors aligned with an eigenvector of $\boldsymbol{Q} = \langle \boldsymbol{x}\boldsymbol{x}^\top \rangle$. Only the eigenvector corresponding to the largest eigenvalue is stable, the rest are saddle points.

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For Oja's rule with a linear neuron, the stable fixed point is the unit vector $\hat{\mathbf{w}}$ aligned with the largest-eigenvalue eigenvector of $\boldsymbol{Q} = \langle \boldsymbol{x}\boldsymbol{x}^\top \rangle$.

Summary

Summary

LTP allows neuronal activity to drive lasting changes (memories)

Hebb proposed the idea that correlated neuronal activity drives synaptic potentiation

NMDA detect coincident presynaptic [spikes/glutamate] and postsynaptic depolarize

STDP is a form of Hebbian plasticity sensitive to temporal correlations

Without addititional mechanisms, the Hebbian rule Δweight $\propto$ post $\cdot$ pre is UNSTABLE

Two stabilized Hebbian rules include Oja and BCM rule.

For a linear neuron, the fixed points of Oja's rule align with a PRINCIPAL COMPONENT of the data's second moment $Q = \langle xx^\top \rangle$.

! Homework Exercise: Viewing a learning rule as a continuous-time, averaged weight update in the form of an ODE can simplify analysis of weight evolution. Consider Oja's rule for a linear neuron with just a single input x, which has nonzero mean and variance. Find the fixed points, and show which ones are stable or unstable.

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end

Note:

We don't have time to cover these types of learning, but students interesting in pursuing this area further should look up:

- Behavioural-Timescale Synaptic Plasticity (BTSP)
- Three-factor learning
- Temporal difference learning
- Classical conditioning and the Rescorla–Wagner model
- Reinforcement learning / Q-learning
- Predictive coding

end

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